Chapter 1 Propositional Logic

Chapter 2 First Order Logic

2.0.1 Mathematic Structure

Definition 2.0.1.1 A **mathematic structure** consists of following elements:

- Universe A;
- Relations $R_1, \dots$;
- Functions $F_1, \dots$, where

$$F_i : A^{k_i} \to A$$

- Constants $a_1, \dots$;

We will take the following notion

$$\mathfrak{A} := (A, R_1, \dots; F_1, \dots; a_1, \dots)$$

We will refer to some structures: The details will not be listed here.

- Ordering Relations: Partial Ordering;Linear Ordering; Strict Partial Ordering; Total/Connected Ordering;
- Algebraic: Groups, Rings and Fields;
- Universe of Partial Order Set;

2.0.2 First-Order Language

Definition 2.0.2.1 A **first-Order Language** L is specified by: Symbols:

- **Nonlogical Symbols**
  - L-Relation symbols Rs_L;
  - L-Function symbols Fs_L;
  - L-Constant Symbols Cs_L;

  Equipped with a function

$$v_L : Rs_L \cup Fs_L \to \omega - \{0\}$$

- **Logical Symbols**
  - Variables: A denumerable set $Vb = \{v_n | n \in \omega\}$;
  - The equality symbols $\doteq$;
  - The connectives $\neg, \vee, \wedge, \to, \leftarrow$;
  - The quantifiers $\exists, \forall$;

2.0.3 Syntax

Definition 2.0.3.1

- Terms:
  - –
  - –
- Formulas:
  - –
  - –

♠ **Proposition 2.0.3.2** (Unique Readability)

- **Proof** ... □

♠ **Proposition 2.0.3.3** (Terms Recusion Theorem)

- **Proof** ... □

♠ **Proposition 2.0.3.4** (Formula Recursion Theorems)

- **Proof** ... □

2.0.4 Structures

Definition 2.0.4.1 (Structure)

2.0.5 Semantics

2.0.6 Metatheory of FOL

Chapter 3 Modal Logic

Chapter 4 Deductive Systems

4.1 Hilbert-Style Systems

4.2 Natural Deductions

4.3 Sequent Calculus

Gentzen LK

Gentzen LJ

Chapter 5 Gödel's Theorems

Chapter 6 Second-Order System